

Introduction

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Lecture 1

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1 Introduction

1.1 Mathematical Programming

- A way to use math to make the best possible decision when there are rules to follow (constraints)
- You have choices you can control (like how much of each product to make)
- You have a goal (like maximizing profits or minimizing cost)
- You have constraints (limits such as time, money or resources)
- Math helps you find the best solution that satisfies all the limits

1.2 Production planning model for Orkla

$$\begin{aligned}
\text{Total cost} &= \sum_{t=1}^W \sum_{i=1}^N \sum_{j=1}^N C_{\text{real}} \times T_{ijt} + \sum_{t=1}^W \sum_{i=1}^N h_{\text{real}} \times I_{\text{real}} \\
&= \sum_{t=1}^W \sum_{i=1}^N \sum_{j=1}^N s_{\text{real}} \times \text{unit labor cost} \times \# \text{ of operators} \times T_{ijt} \\
&\quad + \sum_{t=1}^W \sum_{i=1}^N h_{\text{real}} \times I_{\text{model}} \times \text{capacity} / \text{machine time} \\
&= \sum_{t=1}^W \sum_{i=1}^N \sum_{j=1}^N s_{\text{model}} \times \text{capacity} \times \text{unit labor cost} \times \# \text{ of operators} \times T_{ijt} \\
&\quad + \sum_{t=1}^W \sum_{i=1}^N (h_{\text{real}}) \times \text{capacity} / \text{machine time} \times I_{\text{model}} \\
&= \sum_{t=1}^W \sum_{i=1}^N \sum_{j=1}^N C_{\text{model}} \times T_{ijt} + \sum_{t=1}^W \sum_{i=1}^N h_{\text{model}} \times I_{\text{model}} \\
&\quad \text{s.t.}
\end{aligned}$$

$$I_{i,t-1} + X_{it} - I_{it} = D_{it} \quad \forall i, t \quad (\text{to balance the inventory}) \quad (1)$$

$$X_{it} - Y_{it} \leq 0 \quad \forall i, t \quad (\text{to ensure whenever } X_{it} > 0 \text{ then } Y_{it} = 1) \quad (2)$$

$$\sum_{i=1}^N X_{it} + \sum_{i=1}^N \sum_{j=1}^N T_{ijt} s_{ij} \leq 1 \quad \forall t \quad (\text{capacity constraint}) \quad (3)$$

$$Y_{it} \leq \omega_t \quad \forall i, t \quad (\omega_t = 1 \text{ if any item is produced in period } t) \quad (4)$$

$$\sum_{i=1}^N Y_{it} - 1 \leq (N - 1) \delta_t \quad \forall t \quad (5)$$

$$\omega_t \leq \sum_{i=1}^N \alpha_t \leq 1 \quad \forall t \quad (6)$$

$$\omega_t \leq \sum_{i=1}^N \beta_t \leq 1 \quad \forall t \quad (7)$$

$$\alpha_{it} \leq Y_{it} \quad \forall i, t \quad (8)$$

$$\beta_{it} \leq Y_{it} \quad \forall i, t \quad (9)$$

- Constraint (4) to (10) determine the first and last product to be produced in a period, and also find the product for which the machine is setup at the end of each period
- (6) and (7) Ensure that $\alpha \wedge \beta = 1$ for exactly one product in a given period if production occurs in that period
- (8) and (9) ensure that in a no-production period, all $\alpha = \beta = 0$

$$\sum_{j=1}^N \gamma_{jt} = 1 \quad \forall t \quad (\text{only one product can be set up at the end of each period}) \quad (10)$$

$$\sum_{j=1}^N T_{ijt} \geq Y_{it} - \alpha_{it} \quad \forall i, t \quad (11)$$

$$\sum_{j=1}^N T_{ijt} \geq Y_{it} - \beta_{it} \quad \forall i, t \quad (12)$$

$$T_{ijt} \geq \gamma_{i,t-1} - \omega_t - 1 \quad \forall i \neq j, t \quad (13)$$

- (11) and (12): Apply when ≥ 2 products are produced in a period: force at least one T_{ijt} to be 1 for each product i , except when this product is either the first or the last product in the sequence.
- (13) is to ensure a setup when period t is production-free and the setup state at the end of $t - 1$ is different from the setup state at the end of t
- (14) and (15) counts setups between periods in which the machine is not idle

$$T_{ijt} \geq \alpha_{it} + \gamma_{j,t-1} \quad \forall i \neq j, t \quad (14)$$

$$T_{ijt} \geq \alpha_{it} + \beta_{jt} - 1 \quad \forall i \neq j, t \quad (15)$$

$$0 \leq \omega_t \leq 1 \quad \forall t \quad (16)$$

$$X_{it}, I_{it}, \delta_t \geq 0 \quad \forall i, t \quad (17)$$

$$T_{ijt}, Y_{it}, \alpha_{it}, \beta_{it}, \gamma_{it} \in \{0, 1\} \quad \forall i, t \quad (18)$$

$$Y_{i,t} + \gamma_{i,t-1} \geq \sum_{j=1}^N T_{i,j,t} \quad \forall i, t \quad (19)$$

$$Y_{j,t} + \gamma_{j,t} \geq \sum_{i=1}^N T_{i,j,t} \quad \forall j, t \quad (20)$$

$$\beta_{it} + \alpha_{i,t+1} - 1 \leq \gamma_{it} \quad (21)$$

2 A simple optimization problem

A company has one product

- The sales price is 500 per unit
- The cost is 200 per unit
- It takes 5 hours to produce one unit
- There is a total production cap of 750 hours
- The market will buy all units that the company sells

How many units should be produced and sold? What will be the total profit?

$$\begin{aligned} & \max (500 - 200)X \\ & \text{s.t.} \\ & 5X \leq 750 \\ & 0 \leq X \leq \infty \end{aligned}$$

3 A slightly more complex problem

A company has two products, 1 and 2

- $P_1 = 500$ $P_2 = 400$
- $C_1 = 200$ $C_2 = 150$
- $T_1 = 5$ $T_2 = 3$
- There is a total capacity of 750 hours ($5X_1 + 3X_2 \leq 750$)
- Maximum sales of the products are 120 for P_1 and 200 for P_2
 - $X_1 \leq 120$ $X_2 \leq 200$

Okay so basically (without looking at the solution):

$$\begin{aligned} & \max (500 - 200)X_1 + (400 - 150)X_2 \\ & \text{s.t.} \\ & 0 \leq X_1 \leq 120 \\ & 0 \leq X_2 \leq 200 \\ & 5X_1 + 3X_2 \leq 750 \end{aligned}$$

Set-based form

$$\begin{aligned} & \max \left(\sum_i (P_i - C_i)X_i \right) \\ & \text{s.t.} \\ & \sum_i T_i X_i \leq \text{capacity} \quad (\text{here, } 750) \\ & 0 \leq X_i \leq \text{max sales} \quad \forall i \quad (\text{here, } 120 \text{ and } 200) \end{aligned}$$

4 Demand-allocation model

A product is produced at three factories (A , B , and C) and shipped to four markets, (1, 2, 3, 4)

Capacities at the three factories:

A	B	C
100	120	110

Demand at the four markets:

1	2	3	4
55	60	35	50

Cost per unit for shipping:

	1	2	3	4
A	7	4	9	6
B	3	8	5	4
C	6	5	10	5

Formulate a mathematical model to find the optimal quantities to ship from each factory to each market, minimizing total costs

Parameters:

- $c_{i,j}$: Unit cost from factory i to market j
- d_j : Demand at market j
- cap_i : Capacity at factory i

Decision variables:

- $X_{i,j}$: quantity shipped from factory i to market j

$$\begin{aligned}
 & \min \left(\sum_i \sum_j c_{i,j} X_{i,j} \right) \\
 & \text{s.t.} \\
 & \sum_j X_{i,j} \leq \text{cap}_i \quad \forall i \in \{A, B, C\} \\
 & \sum_i X_{i,j} = d_j \quad \forall j \in \{1, 2, 3, 4\} \\
 & X_{j,i} \geq 0
 \end{aligned}$$